maps the data X to the erased representations, Z, which makes use of a helpful observation – $P(X|A = a_0)$ and $P(X|A = a_1)$ are permutations of one another. This erasure function meets the definition of a perfect erasure function. We hypothesize that the fact that the conditional distributions are permutations is essential to the achievement of perfect erasure. This permutation is essential as we can have a bijective map that transforms X into Z without revealing A. We formalize this in the following class of data.

Definition 4.1 (Data Constraints)**.** *We consider the class of data for which the distributions of different groups are permutations of each other:*

$\forall (i, j),$ *(a)* $|\mathcal{X}_i| = |\mathcal{X}_j|,$ *(b)* $\forall x \in \mathcal{X}_i,\ P_i(x) = P_j(\sigma_{ij}(x)),$

where $P_i = P(X|A = a_i)$*, and for which there exists an erasure function* f *that achieves* $I(Z; A) = 0$ *and* $I(X; Z) = H(X|A)$.

The above definition states that the distributions must be *equal up to permutation* and have equal support sizes ($\mathcal{X}_i = \text{supp}(P_i)$). We provide the justification behind the choice of these data constraints in Appendix A.3. Next, we turn to answer the question of how to find such an erasure function f to achieve this definition.

Equal Distributions (*up to permutations*). In the setting, where the distributions P_i's are equal up to permutation, we present the erasure functions. Intuitively, the erasure function is a one-to-one map from each distribution, P_i, to a common distribution, $Q = P(Z)$ (as shown in Figure 2). The common distribution Q can be any permutation of the P_i's. We formalize this in the following method.

Method 4.2 (Perfect Erasure Function)**.** *If the data constraints in Definition 4.1 hold, then define erasure function* f *as:*

$$f(x) = \{\sigma_i(x)|\ x \in \mathcal{X}_i,\ \sigma_i \in \Sigma_i\}, \tag{3}$$

where σ_i *is any bijective function that transforms* P_i *into* Q *defined by the set,* Σ_i *below:*

$$\Sigma_i = \{\sigma : \mathcal{X}_i \to \mathcal{Z} | \forall x \in \mathcal{X}_i,\ P_i(x) = Q(\sigma(x))\}. \tag{4}$$

The above method shows that a piecewise bijective mapping different concept groups, $\mathcal{X}_i$, to a common output support, $\mathcal{Z}$, achieves the outer bounds for perfect erasure (Definition 4.1). The justification is presented in Appendix A.4. Since the choice of Q and σ_i's are not unique, there can be multiple formulations of the erasure function. In the simple scenario, where all P_i's are uniform distributions, then f can be any random bijective map between $\mathcal{X}_i$ and $\mathcal{Z}$.

Unequal distributions. We consider the scenario where the constraints of Definition 4.1 are violated, i.e., the distributions, $\{P_i\}$'s, are not equal (even after permutation). In this setting, achieving the outer bounds mentioned in Definition 4.1 is not possible. However, it is still possible to achieve perfect concept erasure, $I(Z; A) = 0$. Operating with $P(Z) = P(Z|A = a_i)$, we want to find the piecewise map $f_i : \mathcal{X}_i \to \mathcal{Z}$ such that the distribution of the common support is fixed and equal to the conditional distribution, $Q = P(Z) = P(Z|A = a_i)$. Collectively the piecewise components f_i define the erasure function f. Specifically, we seek mapping functions f_i's that maximize the utility, $I(Z; X)$, as shown below:

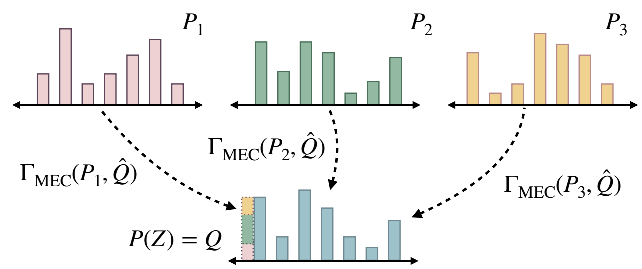

Figure 3: Schematic diagram of the proposed concept erasure process for unequal distributions. Given distributions from different concept groups, $\{P_1, P_2, P_3\}$, we map these representations to a common distribution, Q. Each distribution P_i is mapped to Q using the minimal entropy coupling map, $\Gamma_{\text{MEC}}(P_i, Q)$.

$$\begin{aligned}\max I(Z; X) &= \max I(Z; X|A) \qquad (5)\\ &= \max \sum_i p(a_i) I(Z; X|A = a_i)\\ &= \max \sum_i p(a_i) \left[H(Q) + H(P_i) - H_{\Gamma_i}(Q, P_i)\right]\\ &= \max \left[H(Q) - \sum_i p(a_i) H_{\Gamma_i}(Q, P_i)\right]. \qquad (6)\end{aligned}$$

The optimization of the above objective depends on two key sets of parameters: (i) the distribution of the common support, Q, and (ii) Γ_i's – coupling joint distributions between pairwise distributions (P_i, Q). The second term in the optimization (Eq. 6) is equivalent to estimating the minimum entropy coupling, $H_{\Gamma_i}(P_i, Q)$, which can be obtained efficiently using the greedy algorithm (Section 3) if we know Q. Since Q is unknown, we can simplify the optimization problem as:

$$\begin{aligned}Q^* &= \arg\max_Q J(Q)\\ &= \arg\max_Q \left[H(Q) - \sum_i p(a_i) H_{\min}(P_i, Q)\right]. \qquad (7)\end{aligned}$$

Solving the above objective is difficult and cannot be optimized using standard techniques like gradient descent as we do not know the analytical form of the minimum entropy coupling function, $H_{\min}(\cdot, P_i)$. Even evaluation of the $H_{\min}(\cdot, P_i)$ function is expensive as

MEC estimation is NP-Hard and approximating it still requires running the greedy algorithm. Therefore, we resort to Bayesian Optimization (BO) (Frazier, 2018), which performs optimization by making a small number of queries to the objective $J(Q)$ (Eq. 7). Bayesian Optimization utilizes a surrogate model (usually a Gaussian process (Seeger, 2004)) to fit the observed query points. Using an acquisition function, BO explores different areas of the input space to maximize the objective. We will refer to the solution returned by Bayesian Optimization as $\hat{Q}_{\text{BO}}$; however, it is not guaranteed to be the optimal solution.

We hypothesize that the set of input distributions $\bar{P}$ are a good candidate for output distribution Q as they are stationary points for the objective $J(Q)$. The best achievable utility, $I(Z;X)$, when $Q \in \overline{P}$ is shown by the point in Figure 1. The set $\bar{P}$ only represents the set of stationary points and it may be still possible for the global maxima, Q^* to improve upon them. However, when distributions are unequal analytically deriving the global optima $Q^* \notin \bar{P}$ is non-trivial and remains an open question.[2]

To summarize, when underlying distributions P_i's are unequal, we select either by using the stationary points or the solution of Bayesian optimization: $\hat{Q} = \arg\max_{Q \in \bar{P} \cup \hat{Q}_{\text{BO}}} J(Q)$. In practice, we observe that the stationary points $\bar{P}$ often achieve better performance than the BO solution, $\hat{Q}_{\text{BO}}$. Once, we have selected the support distribution $\hat{Q}$, the erasure function is a stochastic map, $f(x) \sim P(Z|X=x)$, that samples the output z from the conditional distribution:

$$P(Z|X=x) = \left\{ \Gamma_i(x) \middle| \; x \in \mathcal{X}_i, \Gamma_i = \arg\min_{\Gamma} H_\Gamma(P_i, \hat{Q}) \right\},$$

where Γ_i is the coupling map corresponding to the minimum entropy coupling between distribution pairs $(P_i, \hat{Q})$. An illustration of this erasure function is shown in Figure 3. Using the result from Lemma 3.3, it is easy to see that for unequal distributions the erasure function doesn't achieve the utility outer bound as $H(X|Z,A) > 0$ (as different $x \in \mathcal{X}_i$ can map to the same $z \in \mathcal{Z}$). Nonetheless, since the output distribution Q is fixed, and MEC yields a valid joint distribution, it still ensures perfect concept erasure, $I(Z;X) = 0$.

We summarize the overall concept erasure algorithm in Algorithm 1. The algorithm is provided with the entire representation set $\mathcal{S} = \{x_1, x_2, \ldots\}$ (where $x_i \in \mathcal{X}$) and the corresponding concept set, $\mathcal{G}$. First, it estimates the distribution of the representations from individual concept groups. Second, it checks whether the underlying distributions are equal up to permutation. Based on this step, the algorithm determines the optimal erasure function. Finally, the selected erasure function is used to generate the erased representations, $\mathcal{O}$.

Algorithm 1 Concept Erasure Using `PEF`

```
Input: Sample Set S = {x_1, x_2, ...}, Concept Set G = {a_1, a_2, ...}.
∀i, P_i = EstimateDistribution(S_i) // (S_i = {x | x ∈ X_i} is the sample set of the i-th concept group)
if ∀(i, j), P_i = σ(P_j) then
    // Distributions are equal up to permutation
    f(x) = {σ_i(x) | x ∈ X_i, σ_i ∈ Σ_i} // (Σ_i is defined in Eq. 4)
    O = f(S)
else
    // Unequal distributions
    Q̂ = argmax_Q [H(Q) − Σ_i p(a_i) H_min(P_i, Q)]
    P(Z|X = x) = {Γ_i(x) | x ∈ X_i, Γ_i = argmin_Γ H_Γ(P_i, Q̂)}
    O = {z ∼ P(Z|X = x) | x ∈ X}
end if
return O // erased representation set
```

5 EXPERIMENTS

In this section, we discuss the experimental setup and evaluation results in detail. The implementation is available here: https://github.com/brcsomnath/PEF.

Baselines. We compare our approach, `PEF`, with several state-of-the-art baselines, including linear concept erasure techniques – INLP (Ravfogel et al., 2020), LEACE (Belrose et al., 2024) and general erasure techniques – FaRM (Chowdhury and Chaturvedi, 2022), KRaM (Chowdhury et al., 2024). We also compare with the state-of-the-art privacy funnel solver that directly optimizes mutual information terms using variational approximation, DCA (Huang and El Gamal, 2024).

Synthetic Data. We conduct experiments in controlled settings using synthetic data, which allows us to directly compare `PEF`'s performance with the theoretical bounds (shown in Figure 1). Otherwise, accurately estimating mutual information (MI) using high-dimensional data is challenging. Specifically, we consider a set of 3D representations sampled from a finite support set. The finite support set is also randomly sampled from a 3D uniform distribution. We use two underlying concepts and sample representations from distinct support for each group. We experiment in two settings where the groups have (i) *equal* and (ii) *unequal* support distributions to simulate the scenarios in Section 4. Finally, we report the MI estimates using $\mathcal{V}$-information (Xu et al., 2019), which uses a classifier to predict the concept (privacy) and original representation (utility) from the representations

[2]Further discussion of optimization in Appendix B.1.

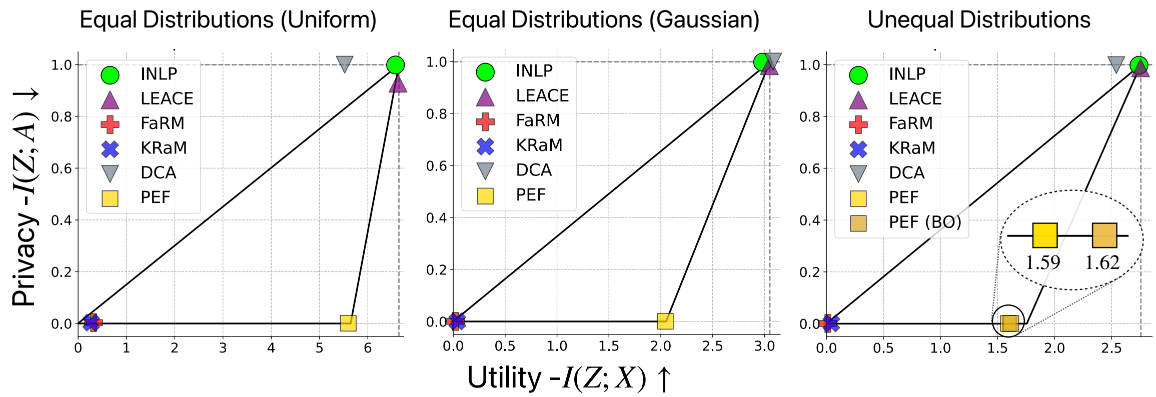

Figure 4: Privacy-Utility tradeoff plots for concept erasure using synthetic data when the underlying representations from different concept groups are sampled from equal uniform (*left*), equal Gaussian (*center*) and unequal (*right*) distributions. We observe that PEF retains significant utility (high $I(Z;X)$) with perfect erasure, $I(Z;A) = 0$.

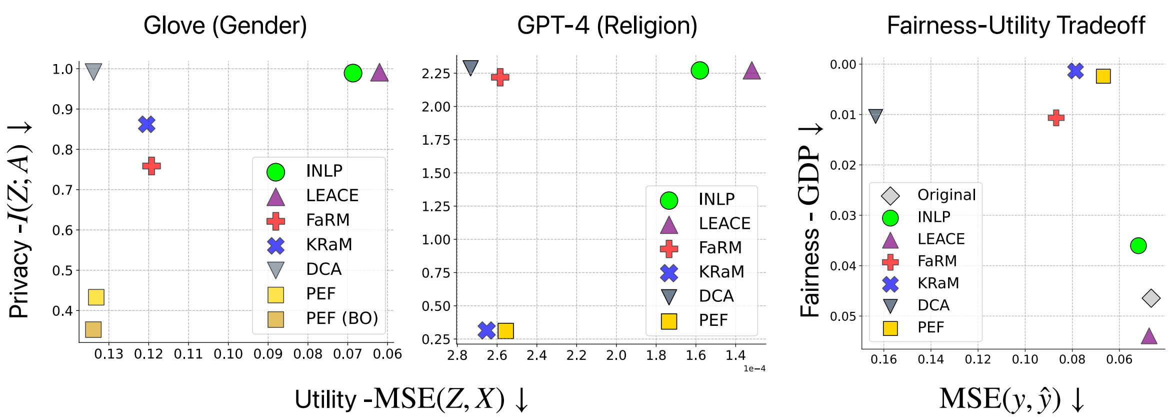

Figure 5: (*Left*) We show the privacy-utility tradeoff for erasing gender information from GloVe embeddings and religion information from GPT-4 embeddings. We observe that PEF achieves significant improvement in privacy compared to baseline approaches. (*Right*) We use the erased GPT-4 embeddings for toxicity classification and plot the fairness-utility tradeoff. We observe that PEF achieves the best fairness-utility tradeoff among the baselines.

post-concept erasure. Since the support of the input representation set is finite, we associate each sample representation with a categorical label. The classifier, used to compute the $\mathcal{V}$-information with the original representations, predicts the label given the erased representation. $\mathcal{V}$-information provides a strong lower bound for the MI estimate when the classifier is constrained by computational resources.

In Figure 4, we report the privacy, $I(Z;A)$, and utility, $I(Z;X)$, tradeoff during concept erasure. The lines describing the funnel structure indicate the theoretical erasure funnel bounds, $\epsilon(u)$, as described in Figure 1. We observe that linear concept erasure approaches, INLP and LEACE, retain significant utility, achieving high $I(Z;X)$ but at the cost of poor privacy, high $I(Z;A)$. DCA performs similar to linear erasure approaches. On the other hand, nonlinear erasure approaches such as FaRM and KRaM erase the concept robustly, but result in a significant drop in utility, low $I(Z;X)$. In contrast to these approaches, PEF achieves high utility while perfectly erasing the concept, $I(Z;A) = 0$. When representations are sampled from equal distributions in Figure 4 (left & center), we observe that PEF achieves the utility outer bound. When the distributions are unequal, in Figure 4 (right), PEF achieves perfect privacy but is unable to achieve the maximum utility, which is expected. In this setting, the Bayesian optimization solution, PEF (BO), slightly improves the utility while achieving perfect erasure.

Real-World Datasets. We evaluate our approach in real-world settings to erase gender from GloVe embeddings and religion from the Jigsaw (jig, 2019) dataset's GPT-4 text embeddings. As these representations are high-dimensional, computing the mutual information with the original representations is challenging. Therefore, we report the mean squared error (MSE) in predicting the original representations from the erased representations using a scikit-learn (Pedregosa et al., 2011) neural network regression model.

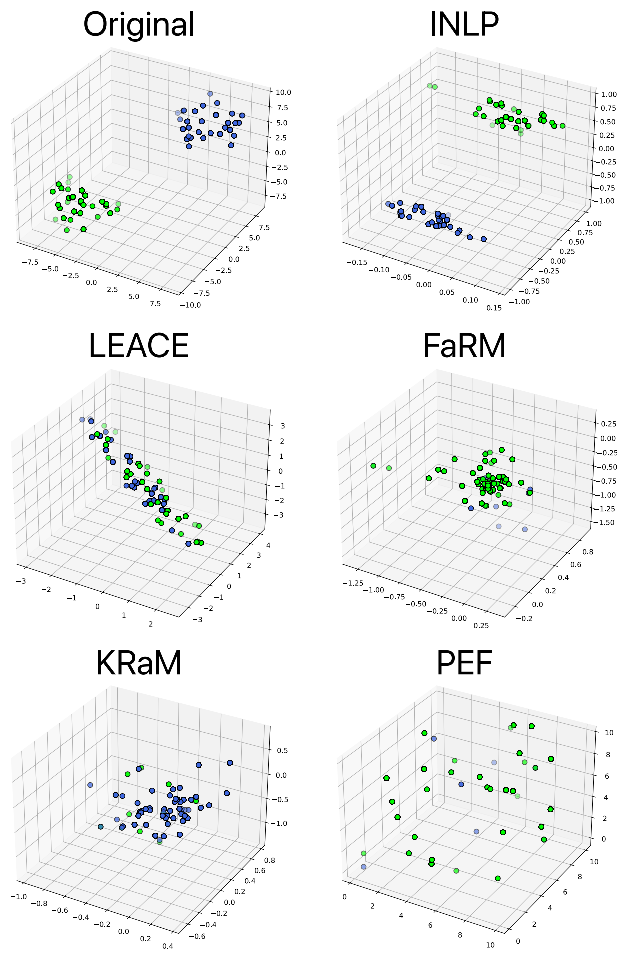

Figure 6: Visualization of the representations post concept erasure using different methods. We observe that existing techniques either leak concept information (INLP & LEACE) or lose utility as representations collapse (FaRM & KRaM). `PEF` can erase concepts properly while maintaining utility.

For Glove embeddings shown in Figure 5 (left), we observe that `PEF` achieves significantly better privacy (low $I(Z;A)$) compared to all baselines while achieving similar utility to methods like FaRM, KRaM, and DCA. We also observe that `PEF` (BO) slightly outperforms `PEF` in terms of privacy. `PEF` assumes access to the original probability distributions. In practice, we have finite high-dimensional samples and use kernel density estimation (Chen, 2017) to estimate the distributions. We hypothesize that the slight drop in utility is due to the unavailability of these original distributions.

In the second setting, we obtain GPT-4 (`text-embedding-3-large`) representations for online comments in the Jigsaw dataset. The objective of this experiment is to erase religion information and utilize the resultant representations for toxicity classification. This ensures that religion does not affect the predicted toxicity scores. We report the results in Figure 5 (center). Similar to the previous setting, we observe significant privacy improvements with similar utility to non-linear erasure techniques (FaRM, KRaM & DCA). In this setting, we found that the local minima $\{P_i\}$' provided a better solution than $\hat{Q}_{\text{BO}}$, therefore, the performance `PEF` and `PEF` (BO) is the same. This is expected as Bayesian optimization is not known to be effective in high dimensions (3072).

After concept erasure, we use the resultant representations to perform toxicity classification. We report the fairness-utility tradeoff, where fairness is computed using generalized demographic parity (Jiang et al., 2021) (for continuous-valued toxicity scores) and utility is the MSE loss of the predictions. In Figure 5 (right), we observe that `PEF` significantly improves fairness GDP scores while having minimal impact on performance (minimal increase in MSE). In general, `PEF` achieves a good balance between utility and fairness compared to the baselines. We provide more details in Appendix C.

Visualization. In this section, we visualize the representations obtained from various methods, as shown in Figure 6. We use synthetic 3D representations, which encode a binary concept shown by representations in blue and green. Post concept erasure, we expect representations from both groups to be indistinguishable. For linear erasure techniques (INLP & LEACE), we observe that the underlying concept can be identified, while non-linear techniques (FaRM & KRaM) achieve better overlap but collapse into a small region of space, signifying a utility loss. In contrast to these approaches, `PEF` shows robust erasure while retaining utility (no collapse observed). We further illustrate the erasure process of LEACE and `PEF` with a simpler example in Appendix C.3.

6 CONCLUSION

In this paper, we study the fundamental limits of perfect concept erasure. Theoretically, we study the maximum amount of information that can be retained while perfectly erasing a concept. Furthermore, we explore the conditions that the underlying data distribution and the analytical form of the erasure function required to achieve perfect erasure. Empirically, we demonstrate the effectiveness of the proposed perfect erasure function (`PEF`) in various synthetic and real-world settings. Although `PEF` can guarantee perfect erasure, it assumes access to a finite support representation distribution, which may be difficult to estimate in low-resource settings. Future works can focus on effectively estimating the underlying distribution using a small number of samples to improve the performance of `PEF`.

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